Database Systems and Information Management Group
Fak. IV Electrical Engineering and Computer Science
Technische Universität Berlin

Thesis Proposal

Thesis Type	Master	**ECTS**	30
Student	Tarik Abu Mukh	**Mat. Number**	0501953
Email	t.abu.mukh@campus.tu-berlin.de	**Primary Advisor**	Lennart Behme
1st Examiner	Prof. Dr. Volker Markl	**2nd Examiner**	Prof. Dr. Matthias Böhm
Academic Program	Computer Science		
Thesis Title	Hardware-Conscious Performance Engineering of Fainder for Distribution-Aware Dataset Search		
Duration Period	From 11/2025 to 5/2026		

1. Introduction / Scientific Background / Related Work

1.1 Specify the *obstacle* to be overcome

Fainder is a recently proposed index for distribution-aware dataset search that evaluates percentile predicates over heterogeneous histograms. The existing prototype implementation of the index, written in Python, incurs significant CPU and memory overhead during index construction and query execution. Because the prototype is not optimized for hardware efficiency (e.g., cache locality, vectorization, and memory management) its performance limits practical use at larger scales. This thesis addresses these inefficiencies by developing a prototype implementation of the Fainder index that is explicitly optimized for modern hardware.

1.2 Specify the *motivation* (Why is this problem interesting?)

Distribution-aware dataset search is increasingly important for data discovery, especially when raw data access is restricted and only metadata, such as histograms, are available. Fainder provides an index for such scenarios, but its current performance prevents deployment in large data repositories (e.g., GitTables with >5M histograms). Improving the efficiency of Fainder's index would enable scalable dataset search engines to support distributional predicates in real-world systems.

1.3 Specify the *novelty* (Was this problem already solved?)

Existing work on Fainder focuses on its algorithmic design and correctness, not its performance engineering. No study has yet explored a low-level, CPU-optimized implementation or analyzed how index layout and query execution affect runtime efficiency.

1.4 Specify the *anticipated impact* (how does solving this problem impact our world?)

Optimizing Fainder will allow researchers and engineers to perform distribution-aware searches at interactive speeds and lower hardware cost, improving data discovery and reuse in scientific, industrial, and open-data settings.

1.5 Specify the *anticipated contributions* (*journal publication*? *open-source software*?)

An open-source, low-level implementation of Fainder, performance benchmarks, and potentially a publication on system-level optimization for data indexing.

1.6 Specify the *scientific background* (or foundational work) that your solution will build upon

This thesis extends Fainder's distribution-aware dataset search by applying hardware-conscious optimization principles from established systems research:

SIMD Vectorization: Vectorized query execution techniques [2,3] using AVX2/AVX-512 may accelerate cumulative density comparisons and bin boundary evaluations in Fainder's percentile predicate processing.

Cache-Conscious Memory Layouts: We will evaluate memory-access optimizations, such as Structure of Arrays (SoA) vs. Array of Structures (AoS), to enhance cache locality during binary search over Fainder's precomputed index. Since memory access is a major bottleneck [8], redesigning Fainder's in-memory representation is essential for better query performance.

Parallel Query Execution: Multicore parallelization strategies from analytical databases may enable parallel processing of percentile predicates across Fainder's cluster-based organization.

Alternative Index Organizations: While Fainder uses sorted arrays with binary search, techniques from tree-based indexes (B-trees, adaptive radix trees [4]) or learned index structures [5] may complement or modify the existing search mechanism to better exploit CPU cache hierarchies.

These represent potential optimization avenues rather than predetermined implementations. The core research question is whether such hardware-conscious techniques can achieve measurable performance improvements for Fainder's unique workload, evaluating percentile predicates over clustered, heterogeneous histogram collections, and which techniques prove most effective.

1.7 Specify the *related work* (e.g., competing solutions corresponding to the research challenge)

Hardware-Optimized Index Structures: Recent research on learned indexes [5], adaptive radix trees [4], and SIMD-accelerated search algorithms [6] demonstrates substantial performance improvements through hardware-aware design. However, these works target traditional point or range queries rather than distribution-aware percentile predicates over heterogeneous histogram collections.

Vectorized Query Processing: Systems like Hyper [2] and recent work on vectorization in analytical databases [3] show the effectiveness of SIMD operations for batch processing. Our work investigates whether similar techniques apply to histogram bin evaluation and cumulative density computation in Fainder's context.

Approximate Histogram-Based Systems: Fainder Approx, histogram sketches (e.g., t-digest, HdrHistogram), and synopsis-based query processing systems provide approximate answers with memory-space trade-offs. However, none focus on low-level CPU optimization for exact percentile evaluation over clustered, heterogeneous histograms.

Performance Gap: No prior work has systematically explored hardware-level optimization of distribution-aware dataset search indexes. This thesis addresses that gap by treating Fainder as an empirical vehicle for understanding how CPU-level design choices affect percentile predicate performance.

2. Goal of the Thesis / Statement of the Research Problem

2.1 State the overarching goal that you aim to achieve in your thesis.

The goal of this thesis is to make Fainder a high-performance, low-overhead system by re-engineering its index and query execution in Rust, leveraging its zero-cost abstractions and memory-safety guarantees [7], with a primary focus on the query phase.
The work aims to improve CPU and memory efficiency through targeted low-level optimizations. Specifically, it will explore cache-efficient memory layouts (SoA vs AoS) to enhance cache locality, apply SIMD vectorization using Rust's SIMD and architecture-specific intrinsics to accelerate percentile computation, employ multicore parallelism via rayon to improve throughput, and evaluate lightweight memory-management strategies such as pooling. Each technique will be quantitatively profiled to understand its impact on latency, cache behavior, and memory footprint.

2.2 Provide a succinct, precise, and unambiguous *statement of the research problem(s)* or *question* to be solved, in order to attain the goal.

How can Fainder's index and query pipeline be redesigned at a low system level to reduce CPU time, memory footprint, and query latency, while maintaining accuracy and scalability across large collections of datasets?

2.3 Specification of the *scope of the thesis*.

2.3.1 State what is *in scope*, particularly, the *subproblems that will be explored.*
The thesis will focus on re-engineering Fainder's query execution phase at a low system level and empirically evaluating hardware-oriented optimizations. The current implementation will first be profiled using perf, valgrind, and cachegrind to identify CPU and memory bottlenecks during percentile evaluation and binary search operations; the core of query-time performance. Based on profiling results, we will introduce targeted optimizations including cache-local data layouts for the pre-computed index, SIMD vectorization of cumulative density comparisons, and multicore parallelization of independent cluster queries. Each optimization will be analyzed through microbenchmarks measuring cache-miss rates, instruction counts, vectorization throughput, and memory-access efficiency. In scope are therefore both the implementation of query-phase optimizations and the experimental assessment of their performance impact, while maintaining exact correctness of query results. While histogram alignment is part of Fainder's index construction phase and outside the primary optimization focus, any query-phase improvements that necessitate changes to the index layout (e.g., restructuring for better cache locality) will be documented and evaluated as part of the construction-time overhead analysis.

2.3.2 State what is *out of scope*.
The thesis will not modify Fainder's algorithmic foundations, mathematical model, or percentile-predicate semantics. It will not explore GPU acceleration, distributed computing, or cloud-based scaling approaches.
Likewise, user-interface design, visualization, and non-performance-related extensions are excluded. The work is confined to single-node, CPU-level optimization and focuses exclusively on execution efficiency. All proposed improvements will therefore remain transparent to the logical behavior of Fainder, ensuring that query results and accuracy remain identical to the reference implementation.

3. Thesis Approach

3.1 Describe the solution approach (e.g., algorithms, data, evaluation metrics, software, systems).
The thesis applies a systems-level performance-engineering methodology to Fainder's query execution. The approach begins with an empirical analysis of the baseline implementation to identify the dominant contributors to query latency and memory usage. Based on these insights, a Rust-based prototype will be developed that incorporates hardware-conscious techniques such as SIMD vectorization, cache-optimized memory layouts, and multicore parallelism. The redesigned components will be evaluated using representative dataset collections under controlled workloads. The analysis focuses on low-level efficiency indicators to determine how each optimization affects Fainder's percentile-predicate evaluation. This method enables a systematic comparison between baseline and optimized variants and provides a quantitative foundation for the ablation study outlined in Section 3.3.2.

3.2 How does the proposed solution differ from the state-of-the-art?
Unlike previous work, which focused solely on Fainder's algorithmic correctness and pruning efficiency, this thesis treats the system as an empirical research object to study how hardware-level

design choices influence distribution-aware query processing. The novelty lies in combining techniques from database indexing, systems programming, and CPU-architecture optimization to analyze their joint impact on percentile-predicate evaluation. Rather than proposing a new indexing algorithm, the thesis investigates the *performance principles* that determine when vectorization, cache layout, and memory-pooling yield measurable gains, thus contributing to a broader understanding of hardware-conscious index design for data discovery systems.

3.3 How will you know if you have *succeeded* in attaining your goal, i.e., solved the research problem(s)?

3.3.1 How will you measure the *effectiveness* of your solution?

Effectiveness will be validated by confirming that all query results remain identical to the baseline Fainder implementation. Correctness will be verified through regression tests on known datasets and statistical equivalence checks between original and optimized outputs.

3.3.2 How and against which baseline(s) will you measure the *efficiency* of your solution?

Efficiency will be measured by comparing the optimized prototype to the original Fainder implementation under identical query workloads. Beyond overall runtime and throughput, the evaluation will record CPU-cycle breakdowns, cache-miss and branch-misprediction rates, vector-instruction utilization, memory-bandwidth usage, and parallel-scaling efficiency across increasing core counts.

Ablation Study: To isolate the contribution of individual optimizations, we will conduct an ablation study where each optimization technique (cache layout restructuring, SIMD vectorization, multicore parallelism) is systematically removed. This will quantify the marginal performance gain of each component and identify which optimizations provide the greatest benefit for Fainder's specific query workload patterns.
These low-level metrics will explain observed performance differences and reveal how each optimization contributes to overall efficiency. Results will be visualized through scalability plots, per-component performance profiles, and ablation comparison charts to provide an interpretable, data-driven explanation of improvements.

4. Implementation Plan and Timeframe

Dates	Milestone	Tasks	Deliverable(s)
01.12 – 31.12	**MS1**	1. Conduct literature review on Fainder, distribution-aware indexing, and CPU/system optimization.	Annotated bibliography, reading notes, and summary of relevant papers.
		2. Summarize findings and identify performance-related gaps	Reading notes and a short gap analysis document.
01.01 – 31.01	**MS2**	3. Gain familiarity with the Fainder source code and datasets.	Reading notes and a short gap analysis document.
		4. Design low-level optimization plan.	Design document with architecture diagrams, performance hypotheses, and planned evaluation metrics.

01.02 – 15.03	**MS3**	5. Implement optimized Fainder components	Functional prototype implementing redesigned index and query logic.
		6. Integrate low-level submodules and bindings	Integrated system build with working C++/Rust bindings and build documentation.
		7. Validate correctness on benchmark datasets.	Validation report comparing results against baseline Fainder output.
16.03 – 15.04	**MS4**	8. Define experimental setup, datasets, and metrics.	Experiment plan describing workloads, datasets, and performance metrics.
		9. Run experiments comparing baseline and optimized versions.	Raw experimental logs, execution summaries, and timing tables
		10. Collect and visualize results.	Performance graphs, scalability plots, and summarized evaluation figures.
16.04 – 30.04	**MS5**	11. Analyze results, interpret anomalies, and discuss scalability trends.	Analysis report with discussion of findings and draft thesis discussion section.
16.04 – 30.04	**TC**[1]	12. complete thesis writeup	submit thesis by the due date[2]
16.04 – 30.04	**SC**[3]	13. Polish and document code, publish in an open-source repository.	publish code in an open repository

Preparing for the Thesis Defense

Students should sign and date that they have read the instructions below.

Date	Tasks	General Instructions
post thesis submission	14. prepare presentation slides	**send slides (as a .pdf file) to advisor(s) and Juan**, no later than 8:00 a.m. (GMT+1), the day of the scheduled defense
defense date	15. deliver the presentation	Time limited: 15' for B.Sc. and 20' for M.Sc.
	16. answer raised questions	ca. 20' for Q&A

Signature: _____ **Date:** _____

[1] thesis complete
[2] add an approximate date, the precise due date will be provided by the Prüfungsamt in writing at a later date.
[3] system complete

5. Bibliography

[1] Behme, L., Galhotra, S., Beedkar, K., and Markl, V. 2024.
Fainder: A Fast and Accurate Index for Distribution-Aware Dataset Search.
Proceedings of the VLDB Endowment 17, 11, 3269–3282.

[2] Neumann, T. 2011.
Efficiently Compiling Efficient Query Plans for Modern Hardware.
Proceedings of the VLDB Endowment 4, 9, 539–550.

[3] Kersten, M., Drebes, A., and Neumann, T. 2018.
Everything You Always Wanted to Know About Vectorized Group-By.
Proceedings of the VLDB Endowment 11, 13, 2168–2180.

[4] Leis, V., Kemper, A., and Neumann, T. 2013.
The Adaptive Radix Tree: ARTful Indexing for Main-Memory Databases.
In Proceedings of the IEEE International Conference on Data Engineering (ICDE), 38–49.

[5] Kraska, T., Beutel, A., Chi, E., Dean, J., and Polyzotis, N. 2018.
The Case for Learned Index Structures.
In Proceedings of the ACM SIGMOD International Conference on Management of Data (SIGMOD), 489–504.

[6] Polychroniou, O. and Ross, K. A. 2014.
A Comprehensive Study of Main-Memory Hash Joins on Multicore CPUs.
In Proceedings of the ACM SIGMOD International Conference on Management of Data (SIGMOD), 369–380.

[7] Matsakis, N. D. and Klock, F. S. 2014.
The Rust Language: Memory Safety Without Garbage Collection.
Communications of the ACM.

[8] Manegold, S., Boncz, P. A., and Kersten, M. L. 2000.
Optimizing Database Architecture for the New Bottleneck: Memory Access.
Proceedings of the VLDB Endowment 9, 3, 266–277.

6. Use of AI Tools for Bachelor's Theses and Master's Theses

Prof. Dr. Volker Markl
Chair of Database Systems and Information Management (DIMA),
Technische Universität Berlin

July 2024

Policy Statement.

1. **Department I Student Services** at TU Berlin recently released **Guidelines for Completing Bachelor's and Master's Theses**[4]. Below the *sixth* guideline (**6.**), as described in the document.

> "*The regulations regarding the use of generative AI tools may vary from faculty to faculty. You should check with your faculty and read the declaration of authorship in section 8 beforehand. As a rule,* ***use is permitted for conducting language checks and improving the language of your own texts or for systematic research****. You should also* ***be aware that AI-generated output may contain errors or be of inferior quality, and that the content needs to be checked and revised****.* ***Examination law requires that the work you submit is your own work and that you disclose any third-party texts/products used****.*
>
> *You can find guidelines on citing generative AI content at https://chicagomanualofstyle.org/qanda/data/faq/topics/Documentation/faq0422.html, https://apastyle.apa.org/blog/how-to-cite-chatgpt, and https://style.mla.org/citing-generative-ai/.*
>
> *To ensure maximum transparency, we recommend documenting communication such as prompts, chat histories, and the use of text taken from chats (for example by copying a chat into a PDF document).* ***Please note that neither copyrighted material nor sensitive information of third parties (e.g., test persons) may be input into the software****."*

Given this update, we wish to clarify some points.

(a) Fak. IV has not issued school-wide regulations concerning the use of generative AI tools.

(b) Per the **Declaration of Authorship** statement outlined in the guidance document (**8.**), students should include a sworn affidavit as the first page of their thesis, bearing the statement below. **Note that willful violation of this declaration may result in legal action.**

> **Declaration of Authorship**
>
> I hereby declare that the thesis submitted is my own, unaided work, completed without any external help. Only the sources and resources listed were used. All passages taken from the sources and aids used, either unchanged or paraphrased, have been marked as such.

[4] Guidelines for Completing Bachelor's and Master's Theses, Technische Universität Berlin, https://www.static.tu.berlin/fileadmin/www/10002136/Lehre/Abschlussarbeit_Hinweise_04_2024_final_eng.pdf

Where generative AI tools were used, I have indicated the product name, manufacturer, the software version used, as well as the respective purpose (e.g., checking and improving language in the texts, systematic research). I am fully responsible for the selection, adoption, and all results of the AI-generated output I use.

I have taken note of the **Principles for Ensuring Good Research Practice** at TU Berlin dated 8 March 2017.

https://www.tu.berlin/en/working-at-tu-berlin/important-documents/guidelinesdirectives/principles-for-ensuring-good-research-practice.

I further declare that I have not submitted the thesis in the same or similar form to any other examination authority.

Berlin, (date)

……

Signature

(c) The general regulation on the usage of AI Tools states: "*… use is permitted when conducting language checks and improving language of your own text*" (e.g., addressing grammar issues) or "*… for systematic research*" (e.g., when conducting literature search[5]).

Tools like *Grammarly* help students to address writing issues (e.g., spelling mistakes, grammar errors) in the written thesis. Moreover, besides the use of AI tools, literature search can still be conducted using **search engines** (e.g., Google), the **ACM Digital Library** (https://dl.acm.org/), **IEEE Xplore** (https://ieeexplore.ieee.org/Xplore/home.jsp), **Google Scholar** (https://scholar.google.com/), as well as our **library** (https://www.tu.berlin/ub), among others.

(d) *"AI-generated output may contain errors or be of inferior quality, and that the content needs to be checked and revised."* Thus, be aware students are ultimately responsible *for* everything included in the thesis, particularly, the correctness of what has been written/submitted.

(e) "***Examination law*** *requires that the* ***work you submit is your own work*** *and that you* ***disclose any third-party texts/products used***." Note there is *"… risk of including text that comes from other sources and [this] could be flagged as plagiarism should proper source references not be included!"*

Be aware that AI tools present a very serious risk of plagiarism. **These tools must be used judiciously, in coordination with your advisor/mentor.** Akin to copying code from *Stack Overflow*, there is a spectrum with respect to when plagiarism occurs. **At DIMA, any thesis with plagiarized content will result in a failure.**

"Limitations and risks of relying on AI-generated content. Generative AI may provide incorrect or false information, contain text from sources that cannot be included without reference, and

[5] The Use of Generative AI for Scientific Literature Searches for Systematic Reviews: ChatGPT and Microsoft Bing AI Performance Evaluation, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11107769/

which may thereby lead to plagiarism, and/or provide unbalanced, non-academic, or incomplete analyses. It is always the user who is responsible for ensuring that everything that is written is correct and in accordance with norms of research ethics. Pay particular attention to the risk of including text that comes from other sources and which would thereby be considered plagiarism without proper source references."

Material adapted/borrowed from "*GUIDELINES FOR USE OF AI-TOOLS IN THE WORK ON MASTER THESES*," The NHH Norwegian School of Economics https://www.nhh.no/en/for-students/masters-thesis/guidelines-for-use-of-ai-tools-in-the-work-on-master-theses/#:~:text=Use%20of%20AI%20tools%20must,work%20on%20the%20master's%20thesis.

(f) Furthermore, "*To ensure maximum transparency, we recommend documenting communication such as prompts, chat histories, and the use of text taken from chats (for example by copying a chat into a PDF document).*"

If AI tools were used in the writing of the thesis, students must: (1) list all AI tools used in the writing process in the **Bibliographic (or References)** section, (2) followed by a **Further Use of AI Tools** section, and (3) an **Appendix: AI Generated Content**. These requirements are discussed further below.

- **Bibliographic** (or **Reference**) entries should follow the guidelines cited here:

  https://chicagomanualofstyle.org/qanda/data/faq/topics/Documentation/faq0422.html
  https://apastyle.apa.org/blog/how-to-cite-chatgpt
  https://style.mla.org/citing-generative-ai/

- In the "**Further Use of AI Tools**" section list which AI Tools were used. Be sure to include the **name and version of the tool** (e.g., ChatGPT 3.5), **the purpose of the tool** (e.g., to organize data, data analysis, idea generation), and **the prompts that were used**.

  For example, "**ChatGPT** was used to generate a research question on …" Another example, "**Grammarly** and **DeepL** were used for grammar, spelling, and stylistic checks. Nearly all of the corrections and suggestions for stylistic improvement made by these AI tools were accepted."

- The screenshots of the prompts that were entered and the response by the chatbot should be included in the **Appendix: AI Generated Content**. The screenshots need to be large enough to allow reading of the text.

  Note: AI generated content should be done *sparingly*, since a key criterion for grading written work is a student's ability to demonstrate critical thinking skills, particularly independent thought. By simply including AI-generated content (e.g.,

copy-pasting), students cannot prove their research skills, since they did not conduct the research and review the research literature themselves.

Material adapted/borrowed from "*Rules for AI Usage in Written Work*," Rheinland-Pfälzische Technische Universität Kaiserslautern-Landau https://ksw.rptu.de/index.php?eID=dumpFile&t=f&f=4956&token=c046e7c9c9884af8497c720a7bef237ce66c636d

2. This policy statement is directed at **students pursuing a Bachelor's or Master's thesis at DIMA** as well as **DIMA thesis advisors**, who should point this policy out to students pursuing a thesis under their guidance.

3. **Advisors**: Please be mindful of this matter and bring any violations of this policy to my attention, so that DIMA will be an environment free of the risk of plagiarism.

4. **Students**: If you are currently pursuing a thesis at DIMA, **sign this form and forward it to Juan Soto via email**, to confirm that you are aware of this policy.

Please sign below to acknowledge that you have read this policy and will adhere to it.

Signature: _____ **Date: _____**